

FINAL EXPLANATION: BIOLOGY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 10 – SUB-STRAND 1.1 CELL STRUCTURE: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why do cells in our body look so different from each other, and how does the shape of a cell determine the job it can do? Use everything you learned in Lessons 1 to 12 to write a complete, evidence-based explanation of The Healing Cut: the Form 1 student in Nairobi who cut a finger while slicing onions for supper. Answer each section in the space under its prompt. Write in full sentences, use correct scientific words, and refer to evidence from our lessons (diagrams, the onion and cheek cell practicals, the specialised cell studies, the sickle-cell micrographs and the cell organisation hierarchy). Where a question asks you to make a claim, give your Claim, your Evidence and your Reasoning. Sketches and labelled diagrams are welcome.

Section 1: Cells as the Starting Point

Think about the cut finger. (a) State the three tenets of cell theory and name the scientist linked to the third tenet. (b) Use cell theory to explain where the new skin cells came from as the finger healed. (c) The bacteria that might infect the cut (for example *E. coli* from badly prepared nyama choma) are prokaryotes, while the body cells fighting them are eukaryotes. Describe two structural differences between these two cell types.

(a) Cell theory has three tenets: all living things are made of cells; the cell is the basic unit of life; and all cells arise from pre-existing cells. Virchow is linked to the third tenet (Schleiden and Schwann contributed the first two ideas). (b) Claim: the new skin cells came from existing skin cells. Evidence: at the cut, epidermal cells divide and migrate to close the surface. Reasoning: because all cells arise from pre-existing cells, new skin cannot appear from nothing; surviving epidermal cells at the wound edge divide to make new cells that seal the wound. (c) Prokaryotes such as *E. coli* have no membrane-bound nucleus (their DNA floats freely in a nucleoid region), whereas eukaryotic body cells have a membrane-bound nucleus. Prokaryotes also have no membrane-bound organelles, whereas eukaryotes do. Prokaryotes are also much smaller (about 1 to 5 micrometres) than eukaryotes (about 10 to 100 micrometres).

Section 2: Plant Cells and Animal Cells

In Lessons 4 to 7 you studied plant cells and animal cells, including onion epidermal cells and your own cheek cells under the microscope. (a) Name the structures that plant cells have

(a) Plant cells uniquely have a cell wall, chloroplasts and a large central vacuole. Animal cells have lysosomes and centrioles, which plant cells lack. Both share a nucleus, mitochondria, ribosomes, endoplasmic reticulum, Golgi apparatus and a cell membrane. (b) The onion cells had a rigid cell wall, giving a regular rectangular shape, and a large central vacuole, which the cheek cells did not have. Both onion and cheek cells had a cell membrane, cytoplasm and a nucleus (stained by iodine in onion and by methylene blue in cheek cells). (c) The onion bulb grows underground and is not exposed to light, so its epidermal cells are specialised for protection and storage, not photosynthesis. Cells build and keep only the structures their job needs, so the absence

but animal cells lack, and the structures animal cells have that plant cells lack. (b) What did you observe in the onion cells that was NOT there in your cheek cells, and what did you see in both? (c) You did not see chloroplasts in the onion epidermal cells. Explain why this is not a mistake. (d) Explain why the lack of a cell wall in animal cells is useful for the cells that close the cut finger.	of chloroplasts is evidence of specialisation. (d) Without a rigid cell wall, animal cells such as epidermal cells can change shape, flatten and migrate. This flexibility lets epidermal cells spread across the wound and close the surface. A rigid wall would stop this movement. (Platelets are cell fragments that gather at the cut to form the clot; they do not migrate to close the skin.)
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Section 3: Specialised Cells and the Healing Cut

Healing the cut involves platelets (cell fragments), neutrophils, macrophages, fibroblasts and epidermal cells. (a) Choose THREE of these. For each, describe one structural feature and explain how it suits the job it does in the healing sequence. (b) Use the sickle-cell anaemia micrographs from Lesson 8 as evidence to write a Claim, Evidence and Reasoning paragraph answering: Does the shape of a cell affect the job it can do? (c) Choose one specialised plant cell (such as a root hair cell, xylem vessel or palisade cell) and explain how its structure suits its function.	(a) Any three of: Platelets are small cell fragments with no nucleus and are sticky, so they aggregate quickly at the cut and start clotting to slow the bleeding. Neutrophils have a multi-lobed nucleus and abundant lysosomes full of digestive enzymes, so they can engulf and digest bacteria. Macrophages have an irregular, flexible membrane, so they can engulf debris and dead cells. Fibroblasts are elongated with extensive rough endoplasmic reticulum and Golgi apparatus, so they can make and secrete collagen to rebuild the tissue. Epidermal cells are tightly packed with an active nucleus, so they divide and migrate to close the skin surface. (b) Claim: yes, cell shape determines the job a cell can do. Evidence: in the micrographs, normal red blood cells are biconcave discs with no nucleus, while the sickle cells are distorted and no longer biconcave. Reasoning: the normal biconcave shape lets red blood cells carry oxygen to the tissues. When the shape is disrupted, as in sickle-cell anaemia, the cells carry oxygen less effectively and function is catastrophically compromised, even though the cells are still alive and contain the same kind of DNA. Because the loss of shape leads to the loss of function, shape is the physical basis of function. (c) A root hair cell has a long extension that increases surface area, so it absorbs water and minerals from the soil more efficiently. (Alternatively: xylem vessels are hollow and lignified, so they can transport water; palisade cells have many chloroplasts, so they carry out photosynthesis.)
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Section 4: Same DNA, Different Cells

The puzzle that started our unit was: if all cells in one organism carry the same DNA, why do they look so different? (a) Use what you learned to answer this puzzle. (b) Use your evidence	(a) All the cells carry the same DNA, but different genes are expressed in different cells. Each cell makes only the structures it needs, so cells develop different shapes and organelles. These structures are suited to each cell's job, which is why a platelet, a neutrophil and a fibroblast look so different. (b) Gene expression is selective, so cells with the same DNA end up with different structures. For example, the onion epidermal cells we observed had a cell wall, vacuole and nucleus but no chloroplasts, while a plant palisade cell has dense chloroplasts. Likewise, a neutrophil with many lysosomes and a fibroblast with extensive rough ER and Golgi are very different. This shows that a cell's structures follow the job it is specialised to do, not simply the DNA it carries.
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from the onion and cheek cell practicals, or the specialised cells you studied, to support your answer to (a). (c) Using the cell organisation hierarchy, explain why no single cell, such as a platelet or a fibroblast, can heal the cut alone.	(c) Cells of the same type form tissues (epidermal cells form epithelial tissue), tissues form organs (the skin), organs form organ systems (the integumentary system), and organ systems form the whole organism. Platelets, neutrophils, macrophages, fibroblasts and epidermal cells each do only one part of the job. Healing needs all of them organised and working together, so shape determines the job at the cell level, but organisation determines survival at the organism level.
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Section 5: Final Explanation of The Healing Cut	
Write a complete explanation answering the driving question: Why do cells in our body look so different from each other, and how does the shape of a cell determine the job it can do? Tell the story of the cut finger in order, from the moment of the cut until the skin is closed. Name at least four cell types or cell fragments involved, describe the shape or organelle that suits each to its job, and show how they work together. Finish by saying how your thinking has changed since Lesson 1.	When the Nairobi student cuts a finger, platelets, which are small, sticky cell fragments with no nucleus, gather at the wound and start clotting, so bleeding slows and a scab forms. Neutrophils, with a multi-lobed nucleus and many lysosomes, engulf and digest bacteria that enter the cut. Macrophages, with irregular, flexible membranes, engulf debris and dead cells. Fibroblasts, which are elongated with lots of rough endoplasmic reticulum and Golgi apparatus, make and secrete collagen to rebuild the tissue. Epidermal cells, which are tightly packed with an active nucleus, divide and migrate across the wound to close the surface, and they can move because animal cells have no rigid cell wall. All the cells came from the same organism and carry the same DNA, but different genes are expressed in each, so each builds different structures. Every cell's shape is a solution to a particular job, so shape determines function. These cells and fragments are organised into tissues, organs (the skin) and organ systems, so they work together as a team and heal the finger so that it looks almost as it did before. At the start I thought healing was one simple process that just happens. Now I understand that it is many jobs done by different specialised cells, each shaped for its job, and organised into a living, self-repairing body.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of cell structure knowledge	Accurately names and describes organelles and structures (cell theory, prokaryote versus eukaryote, plant versus animal cells) and links each to its function with correct scientific vocabulary.	Names most structures and functions correctly, with minor errors or gaps in the comparisons.	Names few structures, or confuses plant and animal features or prokaryotes and eukaryotes.
Linking structure to function	For several specialised cells, clearly explains how a specific structural feature enables the job the cell does, including the sickle-cell evidence.	Explains structure-function links for some cells but the reasoning is sometimes incomplete.	Lists features or functions without connecting them to each other.
Use of evidence and reasoning	Makes clear claims supported by specific	Gives claims with some evidence, but the	Makes statements with little or no

(CER)	evidence from lessons (onion and cheek practicals, micrographs) and reasons logically from evidence to claim.	reasoning is thin or one part of CER is missing.	evidence from the lessons.
Explaining the healing cut and the driving question	Gives a complete, correctly sequenced story of healing, naming at least four cell types or fragments, explaining same DNA with different gene expression, and the tissue-to-organism hierarchy.	Gives a mostly complete explanation, with some cell types, the sequence or the DNA and hierarchy ideas weak.	Gives a partial or unordered explanation that does not answer the driving question.
Reflection and communication	Clearly shows how thinking has changed since Lesson 1, written in organised, clear sentences with correct terms.	Shows some change in thinking, with generally clear writing.	Reflection is missing or very general, and the writing is hard to follow.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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